INCH-POUND
MIL-C-3965F
AMENDMENT 4
15 March 2000
SUPERSEDING
AMENDMENT 3
29 March 1991

MILITARY SPECIFICATION

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-C-3965F, dated 8 June 1972, and is approved for use by all Departments and Agencies of the Department of Defense

PAGE 1

Following preamble, delete inactivation block.

PAGES 3 AND 4

* Delete all reference throughout document to QQ-S-571 and MIL-STD-105. Both documents have been canceled.

PAGE 4

* 2.1 list of military specifications, add:

"MIL-PRF-39006 Capacitors, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, Established Reliability, General Specification For."

PAGE 6

3.13, delete “Temperature cycling" and substitute “Thermal shock.” Make this same change wherever “Temperature cycling” appears in this specification.

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Add the following new paragraphs:

"3.25.3 Established reliability (ER) marking. An ER part manufactured in accordance with MIL-PRF-39006 may be marked and furnished as a non-ER part to this specification, if produced on the same assembly line or lines and provided it is subjected to and meet all the inspection requirements of the ER part."

"3.27 AC ripple life at 85°C (when specified, see 3.1). When tested as specified in 4.6.22, capacitors shall meet the following requirements:

DC leakage (at 85°C and 25°C)	Shall not exceed the applicable value specified (see 3.1).
Capacitance	Shall change not more than ±15 percent from the initial measured value (see 3.9).
Dissipation factor	Shall not exceed the applicable value specified (see 3.1).
Visual examination	There shall be no mechanical damage, obliteration of marking, or leakage of electrolyte."

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TABLE VI, Group IV, add:

"AC ripple life at 85°C (when specified, see 3.1) Extend both brackets to include this test."	3.27	4.6.22

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4.5.1.1, delete and substitute:

"4.5.1.1 Inspection lot. An inspection lot shall consist of one or more styles of capacitors of either tubular foil, foil package, tubular slug, or slug package construction (see 3.1), produced on the same production line or lines, under essentially the same conditions and offered for inspection during a single work week. The sample selected from a lot shall be representative of the styles and voltages in the lot. Hermetically-sealed units shall be kept separate from nonhermetically-sealed units. The following style groupings may be used for quality conformance inspection:

Group 1	-	CL20, CL21, CL23, CL30, CL31, CL32, and CL33 (hermetically-sealed units only).
Group 2	-	CL20, CL21, CL22, CL23, CL30, CL31, CL32, CL33, CL70, CL71, CL72, and CL73 (nonhermetically sealed units only).
Group 3	-	CL10, CL13, CL14, CL16, CL17, CL18, and CL55 (hermetically-sealed units only).
Group 4	-	CL66 and CL67.
Group 5	-	CL51, CL52, CL53, and CL54."

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4.5.2.1.1, line 2: Delete "each month's or each two months'" and substitute "every two months or every four months".

* 4.5.2.1 Group C inspection. Delete in its entirety and substitute: 4.5.2.1 Group C inspection. Group C inspection is not required if the manufacturer is qualified to all equivalent parts for equivalent established reliability styles in MIL-PRF-39006. Group C inspection shall consist of the tests specified on table X in the order shown. Group C inspection shall be made on sample units which have passed group A inspection, except that the units which are to be subjected to tests of subgroup 5 shall be units which have successfully passed subgroup 3 of table IX (see 4.6.1.3.3).

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TABLE X, test column: For subgroups 1, 2, 3, and 4, delete "(every 2 months)" and substitute "(every 6 months)" For subgroups 5 and 6, delete "(every 4 months)" and substitute "(every 12 months)" (This alters a change made by amendment 2).

Subgroup 3, add:

"AC ripple life at 85°C (when specified, see 3.1)	3.27	4.6.22"

Insert a bracket before "6".

PAGE 20

4.6.10, line 2: Delete "102" and substitute "107".

4.6.10(b): Delete "D" and substitute "A".

PAGE 25

Add the following new paragraph:

"4.6.22 AC ripple life at 85°C (when specified, see 3.1). Capacitors shall be tested as specified in 4.6.19.1, with the following exceptions:

(a) Operating conditions - Capacitors shall be tested at 40 ±2 kHz sinusoidal; current shall be as specified (see 3.1).

(b) Polarizing voltage shall be one-half the rated dc voltage."

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* "6.6.1, delete and substitute: 6.6.1 Retention of qualification and conformance inspection. For the purpose of retention of qualification and conformance inspection (see 4.4.4 and 4.6), test data on identical items covered by MIL-PRF-39006 may be used."

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